A Short Note on Energy Decay for the Heat Equation

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Abstract. We show that solutions of the heat equation on a bounded interval with homogeneous Dirichlet data lose energy exponentially fast. The proof only uses integration by parts and the Poincaré inequality.

Setting

Let $u(x,t)$ solve

$$\partial_t u = \nu \partial_x^2 u, \quad 0 < x < L, \quad u(0,t) = u(L,t) = 0, \tag{1}$$

with $\nu > 0$. We define the energy

$$E(t) = \frac{1}{2} \int_0^L u(x,t)^2 \, \mathrm{d}x. \tag{2}$$

The energy identity

Multiplying Equation 1 by u and integrating in x gives

$$\frac{\mathrm{d}E}{\mathrm{d}t} = \nu \int_0^L u \, \partial_x^2 u \, \mathrm{d}x = -\nu \int_0^L (\partial_x u)^2 \, \mathrm{d}x, \tag{3}$$

where the boundary term $[u \, \partial_x u]_0^L$ vanishes because of the Dirichlet data.

Exponential decay

The Poincaré inequality $\int_0^L (\partial_x u)^2 \geq (\pi/L)^2 \int_0^L u^2$ turns Equation 3 into $\mathrm{d}E/\,\mathrm{d}t \leq -2\nu(\pi/L)^2 E$, hence

$$E(t) \leq E(0) \exp(-2\nu\pi^2 t / L^2). \tag{4}$$

Remarks

The rate in Equation 4 is sharp: it is attained by $u = e^{-\nu\pi^2 t/L^2}\sin(\pi x/L)$, the first eigenfunction of $-\partial_x^2$ with Dirichlet conditions.

The same argument works in several dimensions, replacing ∂_x^2 by the Laplacian and integrating by parts with the divergence theorem.

Nonlinear terms

If one adds a transport term $w\partial_x u$ with $\partial_x w = 0$, the identity Equation 3 is unchanged, because

$$\int_0^L u\, w\partial_x u \,\mathrm{d}x = \frac{w}{2}\big[u^2\big]_0^L = 0. \tag{5}$$

Transport moves energy around but does not create it.